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## Exam questions

# Comprehensive Explanation of Clustering and the k-means Algorithm

### Question

Explain the clustering problem and describe the k-means clustering algorithm. In particular, discuss the function it minimizes and its relationship with a Gaussian model.

## The Fundamental Problem of Clustering

### The Search for Structure in Chaos

Imagine you have a bag of candies of various shapes and colors, randomly mixed. Your brain naturally does this: it **groups** similar candies. Round ones with round ones, square ones with square ones, red ones with red ones.

**Clustering** is exactly this capability, but for data. It's an **unsupervised** method - no one tells you how to group, you must discover the structure by yourself.

### Formal Mathematical Definition

Given a dataset  $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$  with  $\mathbf{x}_i \in \mathbb{R}^D$ , clustering seeks a partition of  $\mathcal{X}$  into  $K$  subsets  $C_1, C_2, \dots, C_K$  such that:

1. **Completeness:**  $\bigcup_{k=1}^K C_k = \mathcal{X}$
  2. **Exclusivity:**  $C_i \cap C_j = \emptyset$  for  $i \neq j$  (for hard clustering)
  3. **Cohesion:** Points in the same cluster are similar
  4. **Separation:** Points from different clusters are dissimilar
- 

## The k-means Algorithm: A Brilliant Idea in its Simplicity

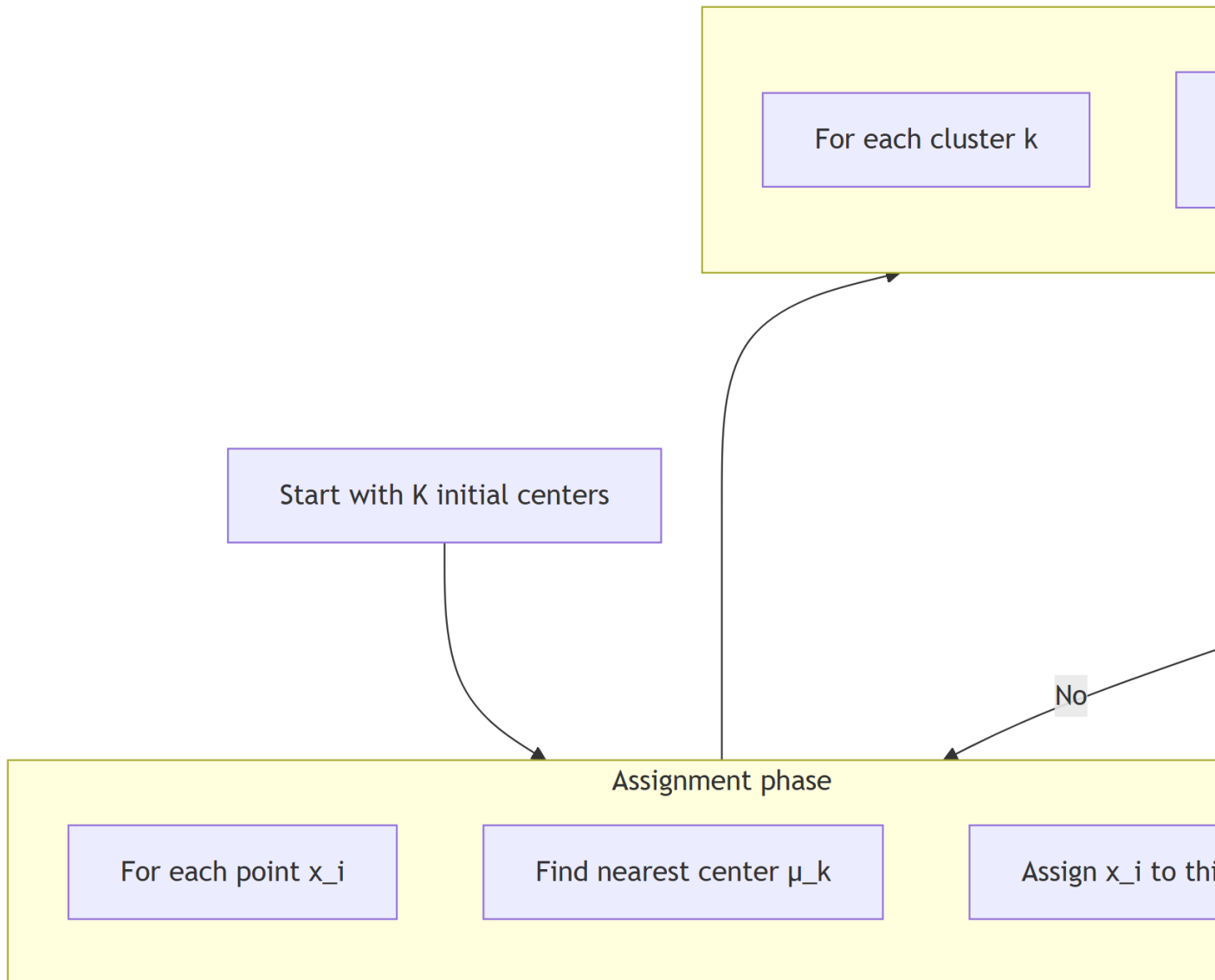
### The Elected Representatives Analogy

Imagine a region with several villages. To reduce travel times, we want to build  $K$  administrative centers. How to place them optimally?

1. **Initialization:** Place  $K$  centers randomly

2. **Assignment:** Each village chooses the nearest center
3. **Readjustment:** Each center moves to the centroid of the villages that chose it
4. **Iteration:** Repeat until stable

This is the essence of **k-means**!



## The Mathematical Heart: The Objective Function

### Precise Formulation

Let:

- **Data:**  $\mathcal{X} = \{\mathbf{x}_i\}_{i=1}^N$ ,  $\mathbf{x}_i \in \mathbb{R}^D$
- **Number of clusters:**  $K$  (fixed a priori)
- **Cluster centers:**  $\mathbf{M} = \{\mu_k\}_{k=1}^K$ ,  $\mu_k \in \mathbb{R}^D$
- **Assignment variables:**  $\mathbf{Z} = \{z_{ik}\}$ , where  $z_{ik} \in \{0, 1\}$  and  $\sum_{k=1}^K z_{ik} = 1$

### The Function to Minimize

k-means minimizes the **sum of squared intra-cluster distances**:

$$J(\mathbf{M}, \mathbf{Z}) = \sum_{i=1}^N \sum_{k=1}^K z_{ik} \|\mathbf{x}_i - \mu_k\|_2^2$$

Where  $\|\cdot\|_2$  is the Euclidean norm:  $\|\mathbf{x}\|_2 = \sqrt{\sum_{d=1}^D x_d^2}$

### Notation Details

| Term                           | Meaning                                                        | Dimension      |
|--------------------------------|----------------------------------------------------------------|----------------|
| $z_{ik}$                       | Binary variable: 1 if $\mathbf{x}_i$ cluster $k$ , 0 otherwise | Scalar         |
| $\mathbf{x}_i$                 | Data point $i$                                                 | $\mathbb{R}^D$ |
| $\mu_k$                        | Center of cluster $k$                                          | $\mathbb{R}^D$ |
| $\ \mathbf{x}_i - \mu_k\ _2^2$ | Squared Euclidean distance                                     | Scalar         |

**Crucial observation:** For a given point  $\mathbf{x}_i$ , only one  $z_{ik}$  equals 1 (the one of the cluster it belongs to), the others equal 0. So for each point, we only count its distance to its own cluster center.

## Optimization by Coordinate Descent

### The Combinatorial Problem

Minimizing  $J$  simultaneously over  $\mathbf{M}$  and  $\mathbf{Z}$  is **NP-hard**: there are  $K^N$  possible assignments!

### Strategy: Alternate Optimization

We use **coordinate descent**:

1. **Fix**  $\mathbf{M}$ , optimize over  $\mathbf{Z}$  (E-step)
2. **Fix**  $\mathbf{Z}$ , optimize over  $\mathbf{M}$  (M-step)
3. **Repeat** until convergence

### E-step: Assignment Optimization

With  $\mathbf{M}$  fixed, the problem becomes:

For each  $i = 1, \dots, N$ , choose  $z_{ik}$  to minimize:

$$\sum_{k=1}^K z_{ik} \|\mathbf{x}_i - \mu_k\|_2^2$$

Under constraint:  $\sum_{k=1}^K z_{ik} = 1, z_{ik} \in \{0, 1\}$

**Solution:** Assign  $\mathbf{x}_i$  to the cluster whose center is closest!

Formally:

$$z_{ik} = \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_i - \mu_j\|_2^2 \\ 0 & \text{otherwise} \end{cases}$$

### M-step: Center Optimization

With  $\mathbf{Z}$  fixed, the problem becomes:

For each  $k = 1, \dots, K$ , find  $\mu_k$  minimizing:

$$\sum_{i=1}^N z_{ik} \|\mathbf{x}_i - \mu_k\|_2^2$$

## Gradient Calculation

$$\nabla_{\mu_k} \left( \sum_{i=1}^N z_{ik} \|\mathbf{x}_i - \mu_k\|_2^2 \right) = -2 \sum_{i=1}^N z_{ik} (\mathbf{x}_i - \mu_k)$$

## Optimality Condition

Setting the gradient to zero:

$$\sum_{i=1}^N z_{ik} (\mathbf{x}_i - \mu_k) = 0$$

$$\Rightarrow \mu_k = \frac{\sum_{i=1}^N z_{ik} \mathbf{x}_i}{\sum_{i=1}^N z_{ik}}$$

**Interpretation:** The optimal center is the **mean** (center of mass) of the points in the cluster.

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## Proof of Convergence

### Monotonic Descent

Let  $J^{(t)} = J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t)})$

1. **E-step:** We choose  $\mathbf{Z}^{(t+1)}$  optimal for fixed  $\mathbf{M}^{(t)}$ :

$$J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t+1)}) \leq J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t)})$$

(strict inequality if at least one assignment changes)

2. **M-step:** We choose  $\mathbf{M}^{(t+1)}$  optimal for fixed  $\mathbf{Z}^{(t+1)}$ :

$$J(\mathbf{M}^{(t+1)}, \mathbf{Z}^{(t+1)}) \leq J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t+1)})$$

(strict inequality if at least one center changes)

3. **Transitivity:**

$$J^{(t+1)} = J(\mathbf{M}^{(t+1)}, \mathbf{Z}^{(t+1)}) \leq J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t+1)}) \leq J(\mathbf{M}^{(t)}, \mathbf{Z}^{(t)}) = J^{(t)}$$

## Consequence

The sequence  $\{J^{(t)}\}_{t=0}^{\infty}$  is **decreasing** and **bounded below** by 0, therefore it converges.

**Warning:** It converges to a **local minimum**, not necessarily global!

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## Deep Relationship with Gaussian Models

### From the EM Algorithm to k-means

To understand the connection, let's start from a **Gaussian Mixture Model** (GMM).

### Gaussian Mixture Model

We assume data is generated by  $K$  Gaussians:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)$$

Where:

- $\pi_k$  is the weight of cluster  $k$  ( $\pi_k \geq 0$ ,  $\sum_k \pi_k = 1$ )
- $\mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)$  is a multivariate Gaussian

### Latent Variables

Let's introduce latent variables  $\mathbf{z}_i = [z_{i1}, \dots, z_{iK}]$  with  $z_{ik} \in \{0, 1\}$  and  $\sum_k z_{ik} = 1$ , indicating which cluster  $\mathbf{x}_i$  belongs to.

### Complete Model

The joint distribution is written:

$$p(\mathbf{x}_i, \mathbf{z}_i) = \prod_{k=1}^K [\pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)]^{z_{ik}}$$

## The EM Algorithm for GMM

EM maximizes the log-likelihood of observed data with respect to parameters  $\theta = \{\pi_k, \mu_k, \Sigma_k\}$ .

### E-step (Expectation)

We calculate the responsibility (posterior):

$$\gamma(z_{ik}) = \frac{\pi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_i | \mu_j, \Sigma_j)}$$

This is a **soft assignment**:  $\gamma(z_{ik}) \in [0, 1]$ ,  $\sum_k \gamma(z_{ik}) = 1$ .

### M-step (Maximization)

We maximize the expectation of the complete log-likelihood:

$$Q(\theta, \theta^{\text{old}}) = \sum_{i=1}^N \sum_{k=1}^K \gamma(z_{ik}) [\log \pi_k + \log \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)]$$

The updates are:

$$\begin{aligned} \mu_k^{\text{new}} &= \frac{\sum_{i=1}^N \gamma(z_{ik}) \mathbf{x}_i}{\sum_{i=1}^N \gamma(z_{ik})} \\ \Sigma_k^{\text{new}} &= \frac{\sum_{i=1}^N \gamma(z_{ik}) (\mathbf{x}_i - \mu_k^{\text{new}})(\mathbf{x}_i - \mu_k^{\text{new}})^\top}{\sum_{i=1}^N \gamma(z_{ik})} \\ \pi_k^{\text{new}} &= \frac{\sum_{i=1}^N \gamma(z_{ik})}{N} \end{aligned}$$



## Transition to k-means: Simplifying Assumptions

### Assumption 1: Spherical and Equal Covariances

Assume  $\Sigma_k = \sigma^2 I$  for all  $k$ , with  $\sigma^2$  fixed and identical for all clusters.

The Gaussian density becomes:

$$\mathcal{N}(\mathbf{x}_i | \mu_k, \sigma^2 I) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|\mathbf{x}_i - \mu_k\|^2}{2\sigma^2}\right)$$

### Assumption 2: Zero Variance Limit

Consider the limit when  $\sigma^2 \rightarrow 0$  (very “peaked” distributions).

In this case,  $\exp\left(-\frac{\|\mathbf{x}_i - \mu_k\|^2}{2\sigma^2}\right)$  decreases very rapidly with distance.

### Assumption 3: Equal Weights

Assume  $\pi_k = 1/K$  for all  $k$  (equiprobable clusters a priori).

## Derivation of the k-means Objective Function

### E-step in Hard Limit

When  $\sigma^2 \rightarrow 0$ , the responsibility  $\gamma(z_{ik})$  becomes:

$$\gamma(z_{ik}) \rightarrow \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_i - \mu_j\|^2 \\ 0 & \text{otherwise} \end{cases}$$

This is exactly a **hard assignment** like in k-means!

## Simplified M-step

With these assumptions, the expectation  $Q$  becomes (up to an additive constant):

$$Q(\theta, \theta^{\text{old}}) \approx -\frac{1}{2\sigma^2} \sum_{i=1}^N \sum_{k=1}^K \gamma(z_{ik}) \|\mathbf{x}_i - \mu_k\|^2 + \text{constant}$$

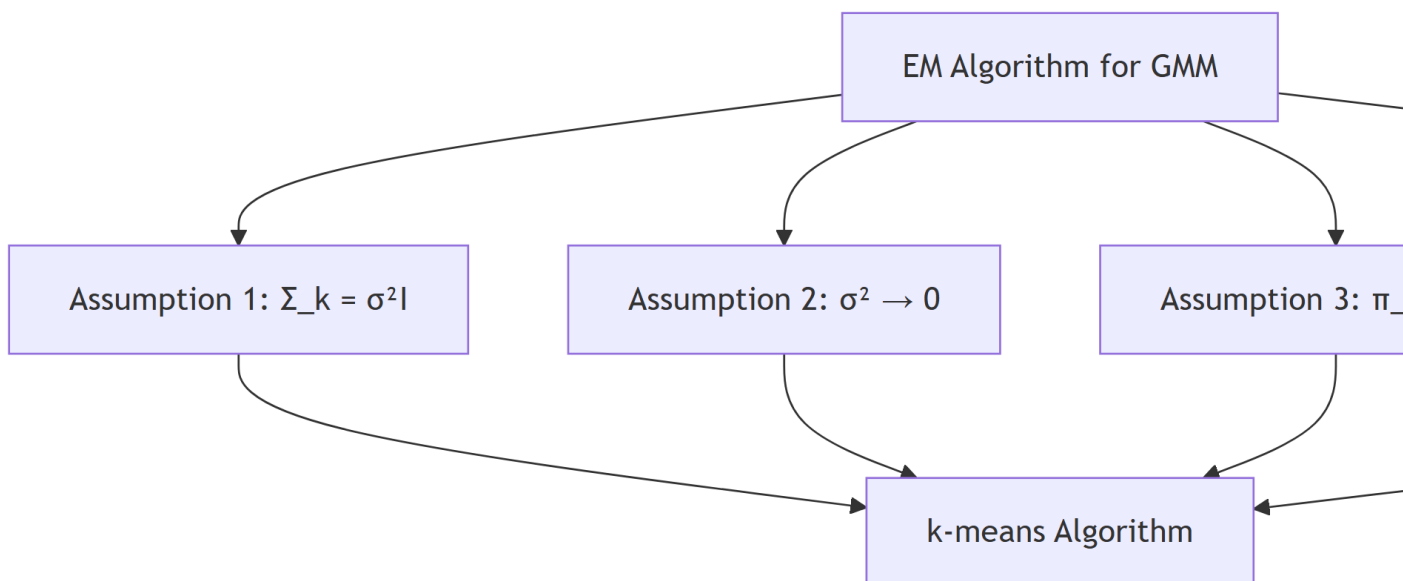
Maximizing  $Q$  is then equivalent to **minimizing**:

$$\sum_{i=1}^N \sum_{k=1}^K \gamma(z_{ik}) \|\mathbf{x}_i - \mu_k\|^2$$

With the hard assignment  $\gamma(z_{ik}) = z_{ik}$ , we recover exactly the function  $J$  of k-means!

## Summary Table: EM $\rightarrow$ k-means

| Aspect                    | General EM for GMM   | k-means (limit case)     |
|---------------------------|----------------------|--------------------------|
| <b>Assignment</b>         | Soft (probabilistic) | Hard (deterministic)     |
| <b>Covariances</b>        | Arbitrary $\Sigma_k$ | $\Sigma_k = \sigma^2 I$  |
| <b>Variance</b>           | Learned from data    | $\sigma^2 \rightarrow 0$ |
| <b>Weights</b>            | Learned $\pi_k$      | $\pi_k = 1/K$            |
| <b>Objective function</b> | Log-likelihood       | Sum of squared distances |



## Practical Implications of This Connection

### What This Teaches Us About k-means

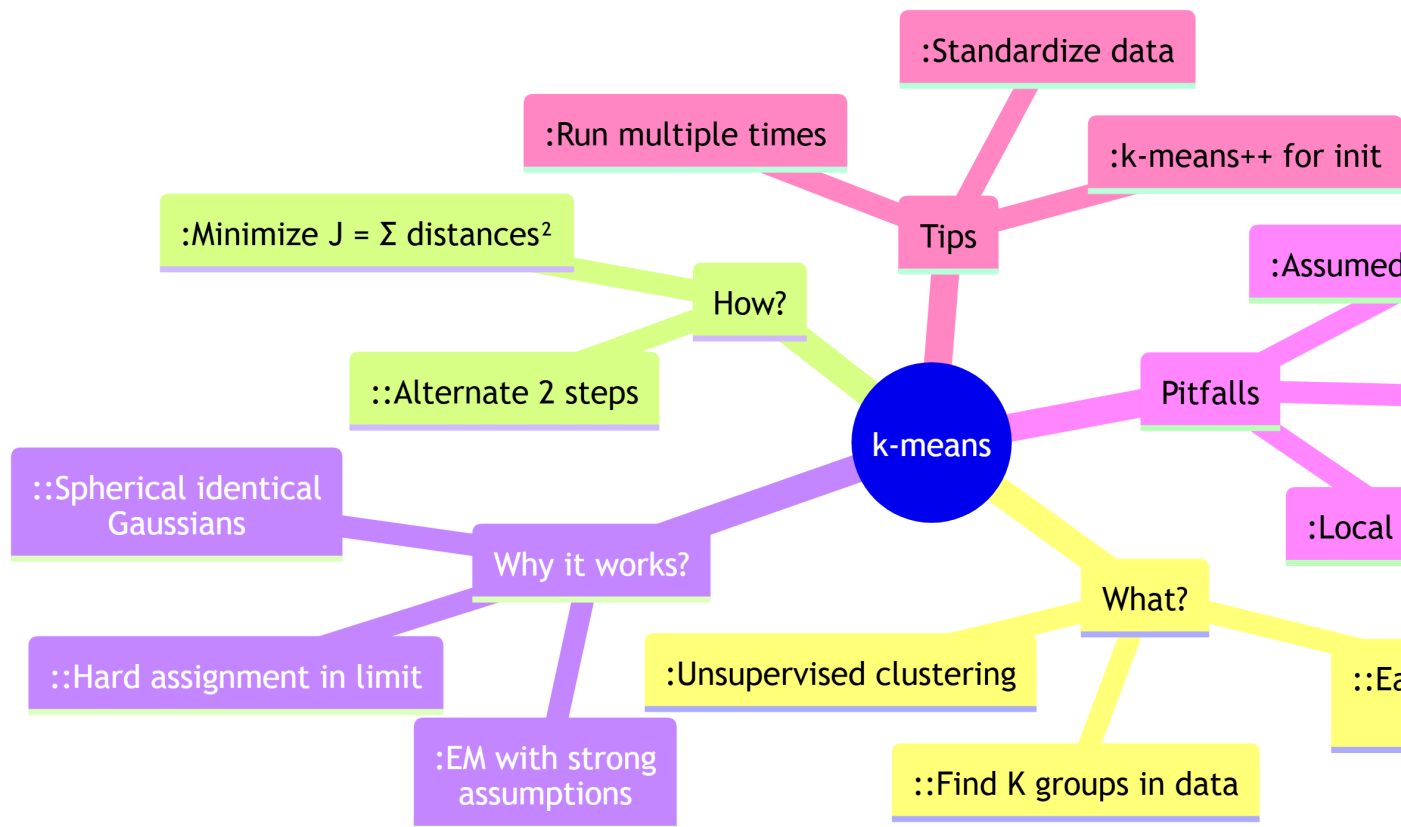
1. **Sphericity assumption:** k-means assumes **spherical** clusters (same variance in all directions)
2. **Scale sensitivity:** Since Gaussians have the same covariance, k-means is sensitive to data normalization
3. **Equal variance assumption:** All clusters are assumed to have the same “spread”
4. **Low overlap limit:** k-means works better when clusters are well separated

### Consequences for Application

- **Normalization necessary:** Always normalize data before applying k-means
  - **Cluster shape:** k-means detects elongated or complex-shaped clusters poorly
  - **Choice of K:** More difficult when clusters have very different variances
-

## Final Intuitive Version

### The Essence in One Image



### Key Takeaway

**k-means is a special case of EM** where we assume each cluster is a spherical Gaussian with the same variance, and we take the limit where these Gaussians are infinitely “peaked” (low variance).

This theoretical understanding explains:

- **Why** k-means works so well on certain problems
- **When** to use it (compact, spherical clusters)
- **How** to improve it (good initialization, normalization)

The genius of k-means is having **simplified to the maximum** a complex problem to obtain a fast, simple, and effective algorithm in many practical cases.

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## For the More Curious: Further Reading

### Possible Generalizations

1. **k-medoids**: Uses an actual point as center (less sensitive to outliers)
2. **Fuzzy C-means**: Probabilistic assignment like EM
3. **k-means with different metrics**: Manhattan distance, cosine...

### Open Challenges

- **Automatic choice of K**: Elbow method, information criteria...
- **Clusters with variable densities**: DBSCAN as alternative
- **High dimensions**: Curse of dimensionality

The beauty of k-means lies in its balance between **mathematical simplicity**, **computational efficiency**, and **intuitive interpretability** - a rare and powerful combination in data science.